
Exercises for Chapter 5

- E1. Consider $\mathbb{R}^3$. Is each of the following a subspace? Explain. For each that is a subspace, determine its dimension and give a basis.
- (a) The set of all vectors with all entries positive.
 - (b) The x -axis.
 - (c) The set of all vectors such that the first entry is bigger than the second.
 - (d) The plane containing the points $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.
 - (e) The set of all linear combinations of vectors $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.
- E2. Give an example that shows that the union of two subspaces is not necessarily a subspace.
- E3. Give example(s) that shows that the intersection of two 10-dimensional subspaces of $\mathbb{R}^{20}$ can have dimension anywhere from 0 to 10.
- E4. Complete the following:
- (a) A trio $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ of linearly independent vectors is a basis if and only if for all other vectors $\mathbf{v}$ the collection $\{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{v}\}$ is _____.
 - (b) A trio $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ of linearly independent vectors is a basis if and only if all other vectors $\mathbf{v}$ are in _____ of $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$.
 - (c) A trio $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ of vectors is a basis if and only if every vector $\mathbf{v}$ is _____ of $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$.
- E5. Suppose that $S = \{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is a linearly dependent set. Prove that every vector $\mathbf{v}$ in the span of the set S can be expressed as a linear combination in more than one way.

Some Solutions

- E1. (a) No (e.g. does not contain $\mathbf{0}$)
(b) Yes. Dimension 1. Basis e.g. $(1, 0, 0)$
(c) No (e.g. does not contain $\mathbf{0}$)
(d) No (e.g. does not contain $\mathbf{0}$)
(e) Yes. Dimension 3. Basis e.g. $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.
- E2. Let S be subspace of all multiples of $(1, 0)$ and T be subspace of all multiples of $(0, 1)$; then the union contains $(1, 0)$ and $(0, 1)$ but not their sum $(1, 1)$.
- E3. Consider the vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_{20}$. To get two 10-dimensional subspaces whose intersection has dimension r , for the first subspace take the span of $\{\mathbf{v}_1, \dots, \mathbf{v}_{10}\}$ and for the second subspace take the span of $\{\mathbf{v}_1, \dots, \mathbf{v}_r, \mathbf{v}_{11}, \dots, \mathbf{v}_{20-r}\}$.
- E4. (a) not linearly independent
(b) the span
(c) unique linear combination
- E5. Consider vector $\mathbf{v}$ in the span. This means that $\mathbf{v} = a\mathbf{x} + b\mathbf{y} + c\mathbf{z}$. Since the set is linearly dependent, we have that $\mathbf{0} = d\mathbf{x} + e\mathbf{y} + f\mathbf{z}$ where not all d, e, f are zero. This means that $\mathbf{v} = (a + d)\mathbf{x} + (b + e)\mathbf{y} + (c + f)\mathbf{z}$ as well.